

Problems on Lambda Function

Each problem includes a sample input and expected output for better understanding.

SECTION A - BASIC

Core syntax, built-in operations, and straightforward practice.

1. Write a Python lambda function to return the square of a number.

Input	number = 9
Expected Output	81

2. Write a Python lambda function to return the cube of a number.

Input	number = 4
Expected Output	64

3. Write a Python lambda function to add two numbers.

Input	a = 15 b = 25
Expected Output	40

4. Write a Python lambda function to return the greater of two numbers.

Input	a = 18 b = 27
Expected Output	27

5. Write a Python lambda function to check whether a number is even.

Input	number = 14
Expected Output	True

6. Write a Python lambda function to return Positive, Negative, or Zero based on a number.

Input	number = -8
Expected Output	Negative

7. Write a Python lambda function to return the first character of a string.

Input	text = Python
Expected Output	P

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8. Write a Python lambda function to return the length of a string.

Input	text = LihaTech
Expected Output	8

9. Write a Python lambda function to convert a string to uppercase.

Input	text = python
Expected Output	PYTHON

10. Write a Python lambda function to multiply three numbers.

Input	a = 2 b = 3 c = 4
Expected Output	24

11. Write a Python program to sort a list of tuples by the second value using a lambda function.

Input	items = [(1, 30), (2, 10), (3, 20)]
Expected Output	[(2, 10), (3, 20), (1, 30)]

12. Write a Python program to sort words by their length using a lambda function.

Input	words = ['python', 'AI', 'data', 'cloud']
Expected Output	['AI', 'data', 'cloud', 'python']

13. Write a Python program to square all numbers in a list using map() and lambda.

Input	numbers = [1, 2, 3, 4, 5]
Expected Output	[1, 4, 9, 16, 25]

14. Write a Python program to select all even numbers using filter() and lambda.

Input	numbers = [1, 2, 3, 4, 5, 6]
Expected Output	[2, 4, 6]

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15. Write a Python program to select words beginning with A using filter() and lambda.

Input	words = ['Apple', 'Python', 'AI', 'Data', 'Agent']
Expected Output	['Apple', 'AI', 'Agent']

16. Write a Python program to convert Celsius values to Fahrenheit using map() and lambda.

Input	celsius = [0, 20, 30, 40]
Expected Output	[32.0, 68.0, 86.0, 104.0]

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SECTION B - INTERMEDIATE

Multi-step logic and practical manipulation of the topic.

1. Write a Python program to sort a list of employee dictionaries by age using a lambda key.

Input	employees = [{'name':'Asha','age':29}, {'name':'Ravi','age':24}, {'name':'Neha','age':31}]
Expected Output	['Ravi', 'Asha', 'Neha']

2. Write a Python program to sort a list of names by the last character using a lambda function.

Input	names = ['Asha', 'Ravi', 'Neha', 'Aman']
Expected Output	['Asha', 'Neha', 'Ravi', 'Aman']

3. Write a Python program to apply a 10% discount to every price using map() and lambda.

Input	prices = [500, 1000, 1500, 2000]
Expected Output	[450.0, 900.0, 1350.0, 1800.0]

4. Write a Python program to filter student records with marks of 75 or above using filter() and lambda.

Input	students = [('Asha', 82), ('Ravi', 68), ('Neha', 91), ('Aman', 75)]
Expected Output	[('Asha', 82), ('Neha', 91), ('Aman', 75)]

5. Write a Python program to calculate the product of all numbers using reduce() and lambda.

Input	numbers = [2, 3, 4, 5]
Expected Output	Product = 120

6. Write a Python program to filter palindrome words using lambda.

Input	words = ['level', 'python', 'madam', 'cloud', 'radar']
Expected Output	['level', 'madam', 'radar']

7. Write a Python program to transform each number into a tuple containing the number, square, and cube using map() and lambda.

Input	numbers = [1, 2, 3, 4]
Expected Output	[(1, 1, 1), (2, 4, 8), (3, 9, 27), (4, 16, 64)]

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8. Write a Python program to sort strings without considering uppercase or lowercase using a lambda key.

Input	words = ['banana', 'Apple', 'cherry', 'apricot']
Expected Output	['Apple', 'apricot', 'banana', 'cherry']

9. Write a Python program to find the dictionary key having the maximum value using a lambda key.

Input	scores = {'Asha': 82, 'Ravi': 74, 'Neha': 91}
Expected Output	Neha

10. Write a Python program to sort student records by marks in descending order and then by name in ascending order using lambda.

Input	students = [('Ravi', 85), ('Asha', 92), ('Neha', 85), ('Aman', 92)]
Expected Output	[('Aman', 92), ('Asha', 92), ('Neha', 85), ('Ravi', 85)]

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SECTION C - ADVANCED

Algorithmic problem solving and deeper application of the topic.

1. Write a Python lambda expression to assign grades: A for 90+, B for 75-89, C for 60-74, and D below 60.

Input	marks = [95, 81, 69, 52]
Expected Output	['A', 'B', 'C', 'D']

2. Write a Python lambda expression to calculate tax using slabs: 0% up to 300000, 5% up to 600000, and 10% above 600000.

Input	income = 800000
Expected Output	Tax = 35000.0

3. Write a Python program using filter(), map(), and lambda to return the squares of only even numbers.

Input	numbers = [1, 2, 3, 4, 5, 6, 7, 8]
Expected Output	[4, 16, 36, 64]

4. Write a Python program to sort file-version tuples by major, minor, and patch numbers using a lambda key.

Input	versions = [('app', '1.2.10'), ('app', '1.10.2'), ('app', '1.2.3')]
Expected Output	[('app', '1.2.3'), ('app', '1.2.10'), ('app', '1.10.2')]

5. Write a Python program to sort points by their Manhattan distance from the origin using a lambda key.

Input	points = [(3, 4), (1, 1), (-2, 2), (0, 5)]
Expected Output	[(1, 1), (-2, 2), (0, 5), (3, 4)]

6. Write a Python program to sort ISO date strings chronologically using a lambda key that converts date parts to integers.

Input	dates = ['2026-12-01', '2025-05-10', '2026-01-15']
Expected Output	['2025-05-10', '2026-01-15', '2026-12-01']

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7. Write a Python program to find the longest word using `max()` with a lambda key.

Input	<code>words = ['AI', 'Python', 'MachineLearning', 'Data']</code>
Expected Output	MachineLearning

8. Write a Python program to create a calculator dictionary whose values are lambda functions for add, subtract, multiply, and divide.

Input	<code>operation = 'multiply'</code> <code>a = 7</code> <code>b = 6</code>
Expected Output	Result = 42

9. Write a Python program to sort transaction records by amount descending and, for equal amounts, by transaction ID ascending using lambda.

Input	<code>transactions = [('T3', 500), ('T1', 700), ('T2', 700), ('T4', 300)]</code>
Expected Output	<code>[('T1', 700), ('T2', 700), ('T3', 500), ('T4', 300)]</code>

10. Write a Python program using `reduce()` and lambda to find the largest number without using `max()`.

Input	<code>numbers = [12, 45, 7, 91, 33]</code>
Expected Output	Largest = 91